APPLET

Write a Java program to read the first five lines from a website and display them on the console, and explain what URL Encapsulation is in Java and whether url.openStream() fetches the visible webpage or the HTML code.

Explain what Enterprise JavaBeans (EJBs) are. Describe the different types of EJBs including **Session** Beans, Entity Beans, and Message-Driven Beans**.**

To play Audio:

```
AudioClip clip = getAudioClip(getDocumentBase(), "sound.wav");
clip.play();
```

To Display images:

```
Image img = getImage(getDocumentBase(), "pic.jpg");
g.drawImage(img, 50, 50, this);
```

Example1:

```
import java.applet.Applet;
import java.awt.*;

public class ShapeApplet extends Applet {
   public void paint(Graphics g) {
      // Draw String
      g.drawString("Shapes Example", 50, 50);
```

```
        // Draw Rectangle
        g.drawRect(50, 70, 100, 50);

        // Draw Oval
        g.drawOval(50, 140, 100, 50);

        // Draw Line
        g.drawLine(50, 220, 150, 220);
    }
}
```

HTML

```
<html>
<body>
    <applet code="ShapeApplet.class" width="300" height="300">
    </applet>
</body>
</html>
```

Example 2:

```
import java.applet.Applet;
import java.awt.*;

public class BouncingBall extends Applet implements Runnable {
    int x = 50, y = 50;      // Ball position
    int dx = 2, dy = 2;      // Direction (movement)
```

```
int radius = 20;
Thread t;

public void init() {
    setSize(300, 300);
}

public void start() {
    t = new Thread(this);
    t.start();
}

public void run() {
    while (true) {
        x += dx;
        y += dy;

        // Bounce from walls
        if (x <= 0 || x >= getWidth() - radius)
            dx = -dx;
        if (y <= 0 || y >= getHeight() - radius)
            dy = -dy;
        repaint();

        try {
```

```
            Thread.sleep(20); // speed control
        } catch (Exception e) {}
      }
   }
   public void paint(Graphics g) {
      g.setColor(Color.RED);
      g.fillOval(x, y, radius, radius);
   }
}
```

Example 3:

```
import java.io.*;
import java.net.*;
import java.util.prefs.Preferences;

public class SimpleApp {

   public static void main(String[] args) {
      try {
         // URL Encapsulation - connect to a website
         URL url = new URL("https://www.example.com");
         BufferedReader br = new BufferedReader(new
InputStreamReader(url.openStream()));
         System.out.println("Reading first 3 lines from URL:");
         for (int i = 0; i < 3; i++) {
            String line = br.readLine();
```

```
            if (line == null) break;
            System.out.println(line);
        }
        br.close();

        // Store user preference
        Preferences prefs = Preferences.userRoot().node("SimpleApp");
        prefs.put("username", "Student123");  // save username
        String user = prefs.get("username", "DefaultUser");
        System.out.println("\nSaved Username from Preferences: " + user);

    } catch (Exception e) {
        System.out.println("Error: " + e.getMessage());
    }
  }
}
```

Sockets / Client-Server networking

Server:

```
import java.io.*;
import java.net.*;

public class SimpleServer {
  public static void main(String[] args) throws Exception {
    ServerSocket server = new ServerSocket(5000);
```

```
        System.out.println("Server waiting...");
        Socket s = server.accept();
        BufferedReader br = new BufferedReader(new
InputStreamReader(s.getInputStream()));
        String msg = br.readLine();
        System.out.println("Client says: " + msg);
        s.close();
        server.close();
    }
}
```

Client:

```
import java.io.*;
import java.net.*;

public class SimpleClient {
    public static void main(String[] args) throws Exception {
        Socket s = new Socket("localhost", 5000);
        PrintWriter out = new PrintWriter(s.getOutputStream(), true);
        out.println("Hello Server!");
        s.close();
    }
}
```

Remote Method Invocation

Remote Interface:

```
import java.rmi.Remote;
import java.rmi.RemoteException;

// Remote Interface
public interface Calculator extends Remote {
    int add(int a, int b) throws RemoteException;
    int subtract(int a, int b) throws RemoteException;
}
```

Remote Implementation:

```
import java.rmi.server.UnicastRemoteObject;
import java.rmi.RemoteException;

// Implementation of Calculator interface
public class CalculatorImpl extends UnicastRemoteObject implements Calculator
{
    public CalculatorImpl() throws RemoteException {
        super();// exports the object
    }
    public int add(int a, int b) {
        return a + b;
    }
    public int subtract(int a, int b) {
        return a - b;
    }
}
```

Servlet Example

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class HelloServlet extends HttpServlet {

   // Handles GET requests
   public void doGet(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      out.println("<h1>Hello from GET request!</h1>");
   }

   // Handles POST requests
   public void doPost(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      String name = request.getParameter("username"); // get data from form
      out.println("<h1>Hello " + name + ", this is POST response!</h1>");
   }
}
```